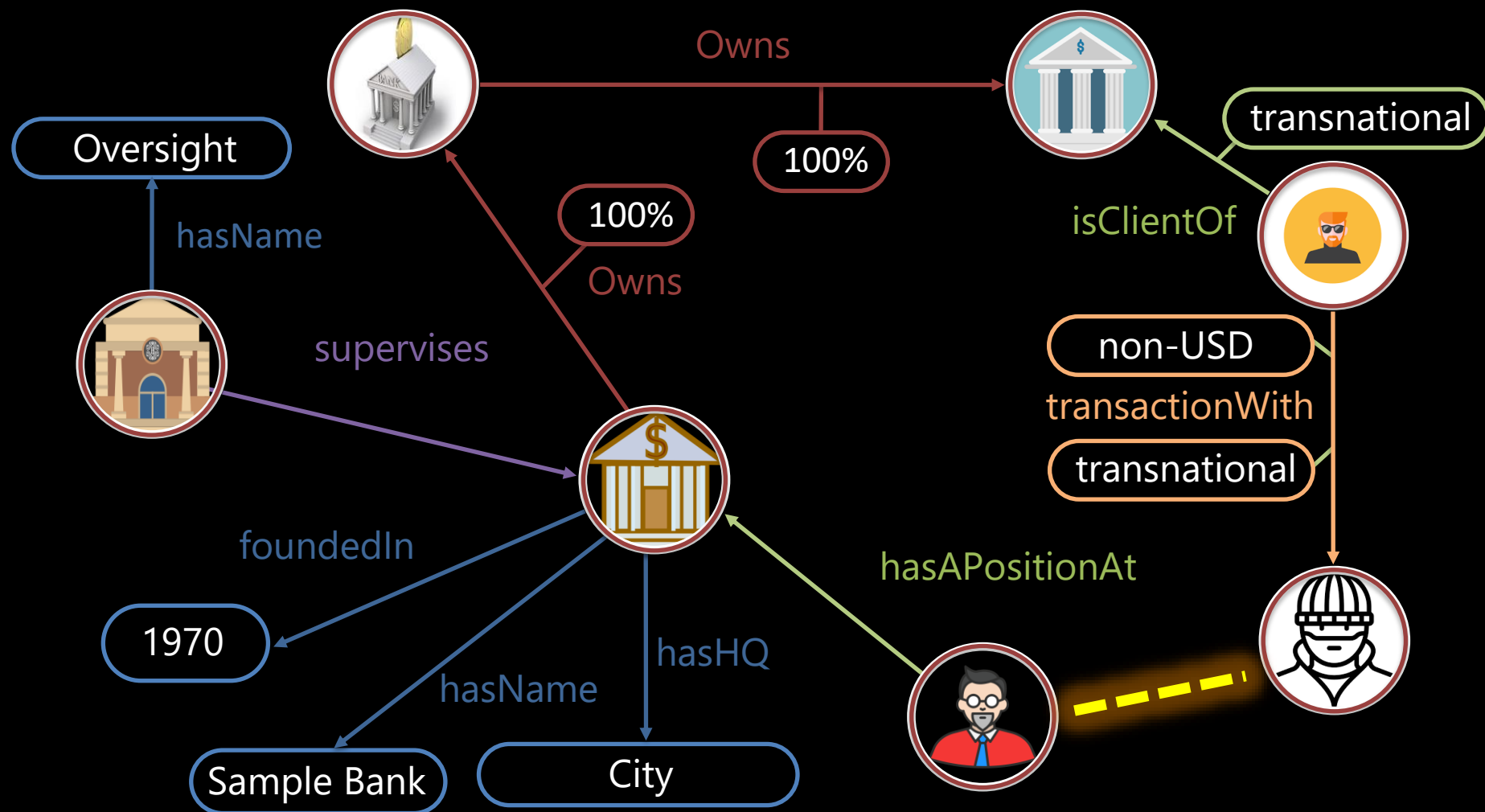


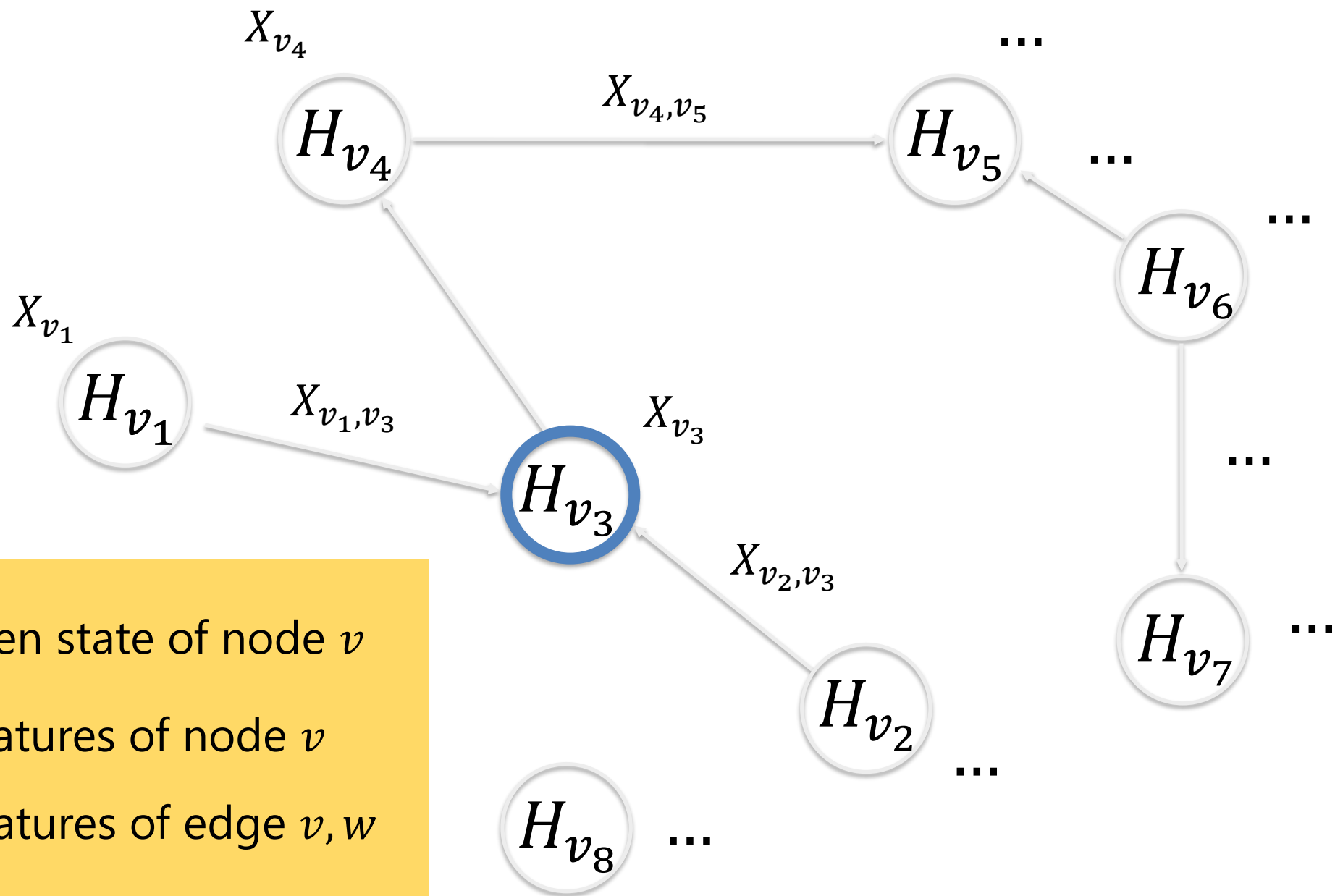


Graph Neural Networks

Message Passing Neural Networks

Emanuel Sallinger

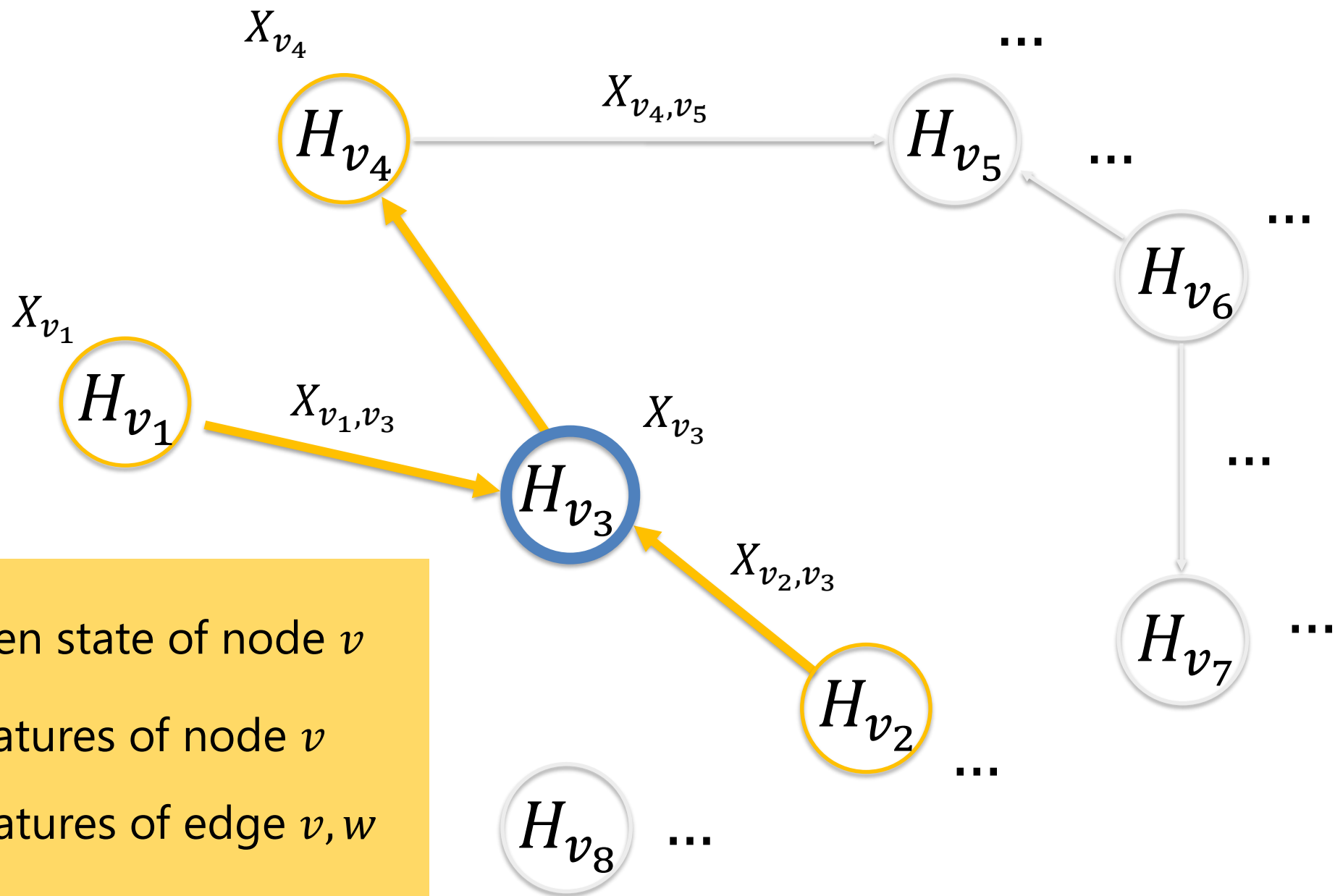




H_v hidden state of node v

X_v features of node v

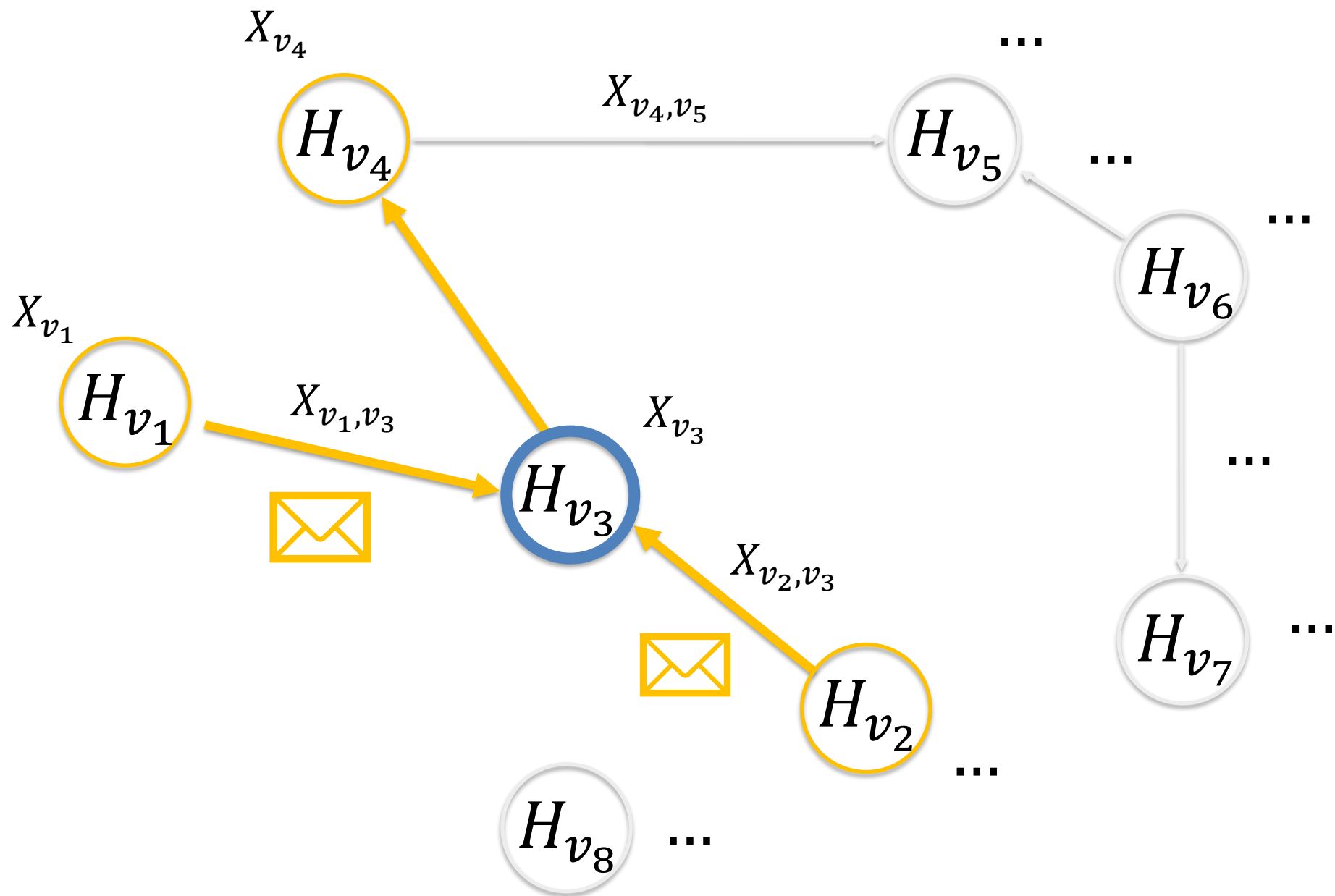
$X_{v,w}$ features of edge v, w

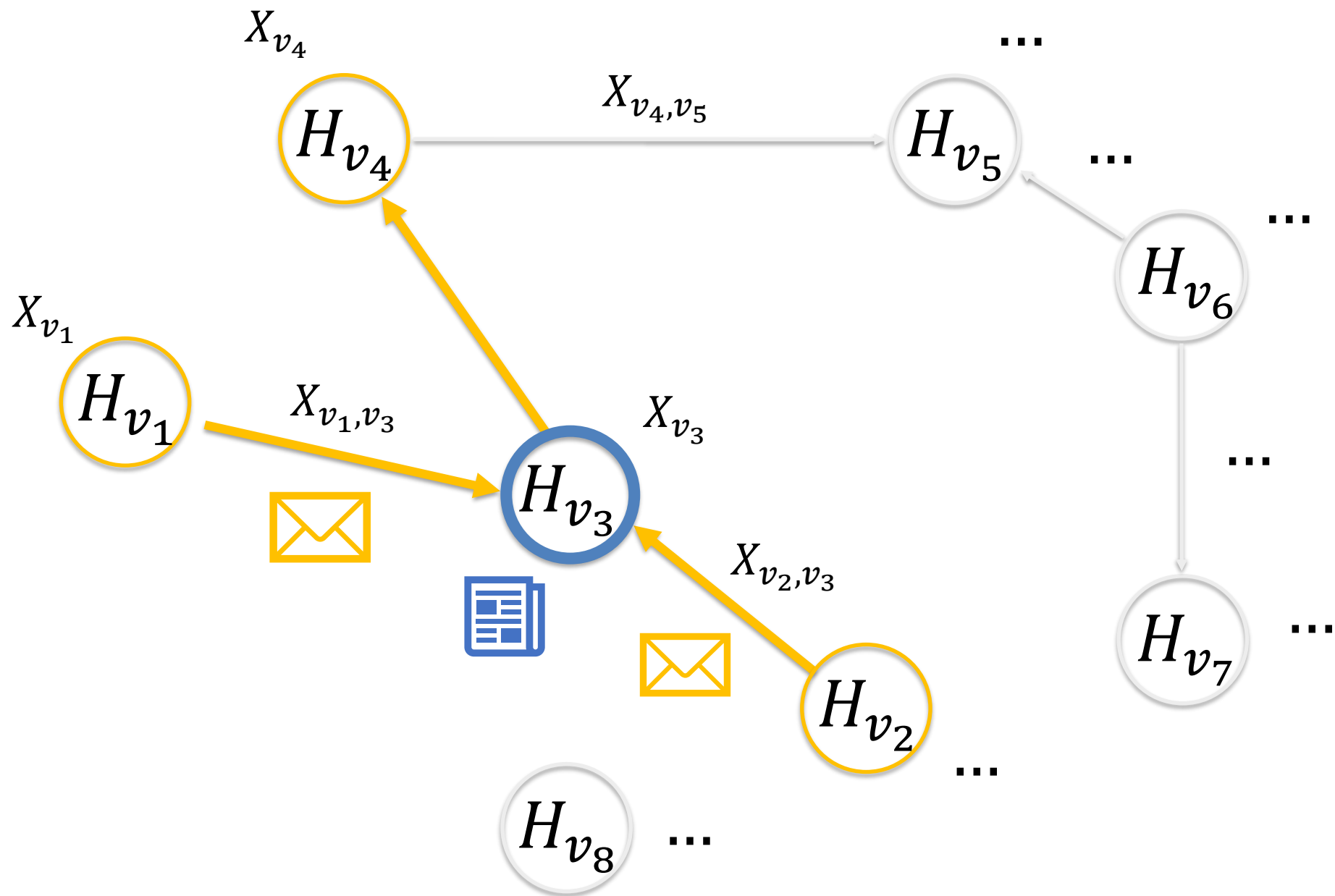


H_v hidden state of node v

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Graph Neural Networks

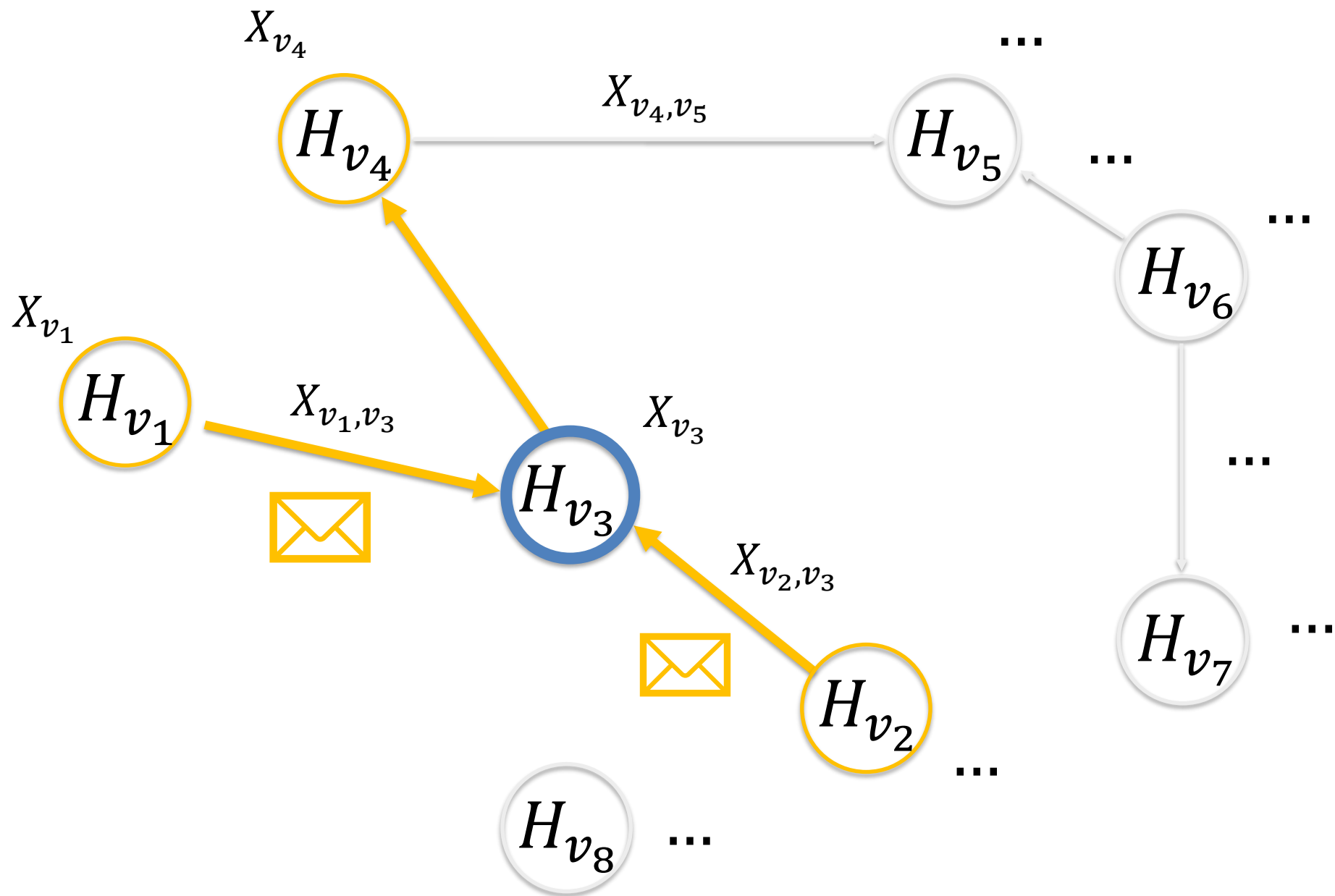
$$\mathbf{H}^{t+1} = F(\mathbf{H}^t, \mathbf{X})$$

F propagation step



Message Passing Neural Networks

$$H_v^{t+1} = u(H_v^t, a_{u \in N(v)}(m(H_v^t, H_u^t, X_{u,v})))$$





Message Passing Neural Networks

messages
from neighbours

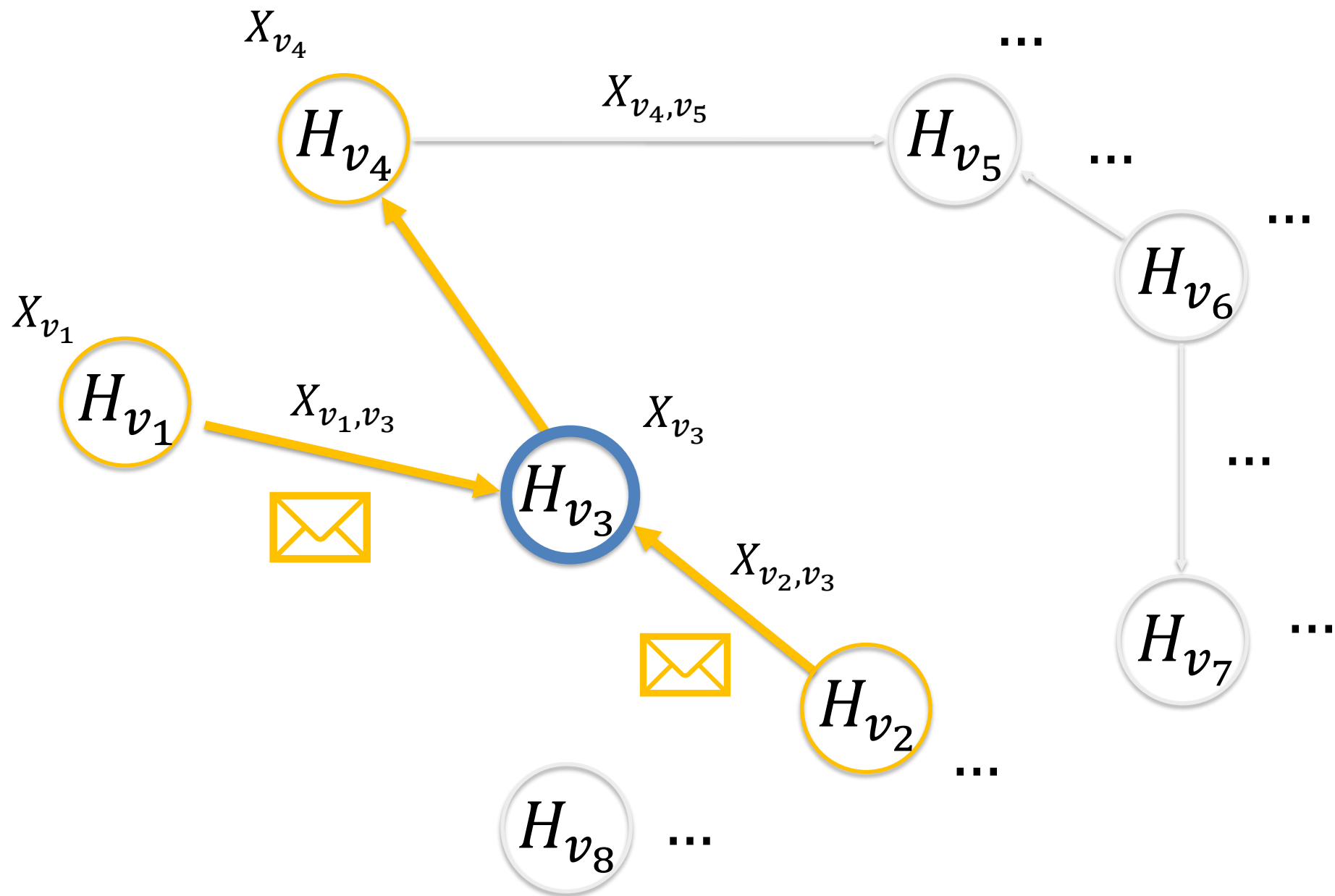


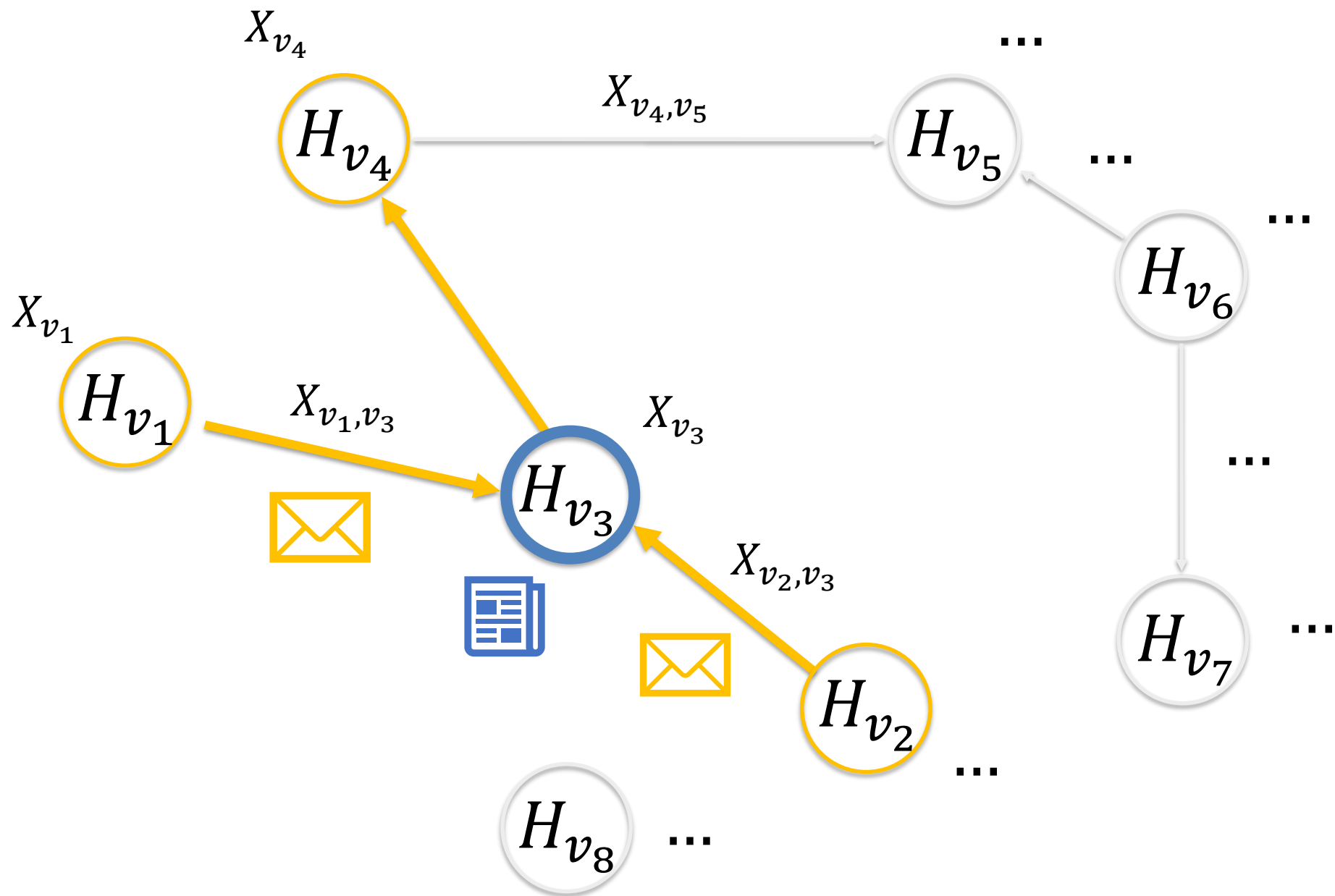
$$H_v^{t+1} = u(H_v^t, a_{u \in N(v)}(m(H_v^t, H_u^t, X_{u,v})))$$



m

message function







Message Passing Neural Networks

aggregate
the messages

messages
from neighbours



$$H_v^{t+1} = u(H_v^t, a_{u \in N(v)}(m(H_v^t, H_u^t, X_{u,v})))$$



m **message** function



a **aggregation** function

$N(v)$ neighbours of v
(e.g., incoming or undirected)



Message Passing Neural Networks



$$H_v^{t+1} = u(H_v^t, a_{u \in N(v)}(m(H_v^t, H_u^t, X_{u,v})))$$



m **message** function



a **aggregation** function

u **update** function

$N(v)$ neighbours of v
(e.g., incoming or undirected)

Neural Message Passing for Quantum Chemistry

Justin Gilmer¹ Samuel S. Schoenholz¹ Patrick F. Riley² Oriol Vinyals³ George E. Dahl¹

Abstract

Supervised learning on molecules has incredible potential to be useful in chemistry, drug discovery, and materials science. Luckily, several promising and closely related neural network models invariant to molecular symmetries have already been described in the literature. These models learn a message passing algorithm and aggregation procedure to compute a function of their entire input graph. At this point, the next step is to find a particularly effective variant of this general approach and apply it to chemical prediction benchmarks until we either solve them or reach the limits of the approach. In this paper, we reformulate existing models into a single common framework we call Message Passing Neural Networks (MPNNs) and explore additional novel variations within this framework. Using MPNNs we demonstrate state of the art results on an important molecular property prediction benchmark; these results are strong enough that we believe future work should focus on datasets with larger molecules or more accurate ground truth labels.

1. Introduction

The past decade has seen remarkable success in the use of deep neural networks to understand and translate natural language (Wu et al., 2016), generate and decode complex audio signals (Hinton et al., 2012), and infer features from real-world images and videos (Krizhevsky et al., 2012). Although chemists have applied machine learning to many problems over the years, predicting the properties of molecules and materials using machine learning (and especially deep learning) is still in its infancy. To date, most research applying machine learning to chemistry tasks (Hansen et al., 2015; Huang & von Lilienfeld, 2016;

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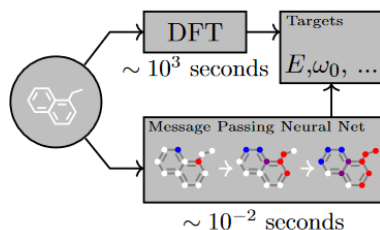


Figure 1. A Message Passing Neural Network predicts quantum properties of an organic molecule by modeling a computationally expensive DFT calculation.

Rupp et al., 2012; Rogers & Hahn, 2010; Montavon et al., 2012; Behler & Parrinello, 2007; Schoenholz et al., 2016) has revolved around feature engineering. While neural networks have been applied in a variety of situations (Merkwirth & Lengauer, 2005; Micheli, 2009; Lusci et al., 2013; Duvenaud et al., 2015), they have yet to become widely adopted. This situation is reminiscent of the state of image models before the broad adoption of convolutional neural networks and is due, in part, to a dearth of empirical evidence that neural architectures with the appropriate inductive bias can be successful in this domain.

Recently, large scale quantum chemistry calculation and molecular dynamics simulations coupled with advances in high throughput experiments have begun to generate data at an unprecedented rate. Most classical techniques do not make effective use of the larger amounts of data that are now available. The time is ripe to apply more powerful and flexible machine learning methods to these problems, assuming we can find models with suitable inductive biases. The symmetries of atomic systems suggest neural networks that operate on graph structured data and are invariant to graph isomorphism might also be appropriate for molecules. Sufficiently successful models could someday help automate challenging chemical search problems in drug discovery or materials science.

In this paper, our goal is to demonstrate effective machine learning models for chemical prediction problems

Deep Tensor Neural Networks

Convolutional Networks for Learning Molecular Fingerprints

Molecular Graph Convolutions

Gated Graph Neural Network

...



Example

update
the state

aggregate
the messages

messages
from neighbours



$$H_v^{t+1} = \text{ReLU} \left(W_1^{t+1} H_v^t + W_2^{t+1} \sum_{u \in N(v)} (H_u^t + b^t) \right)$$



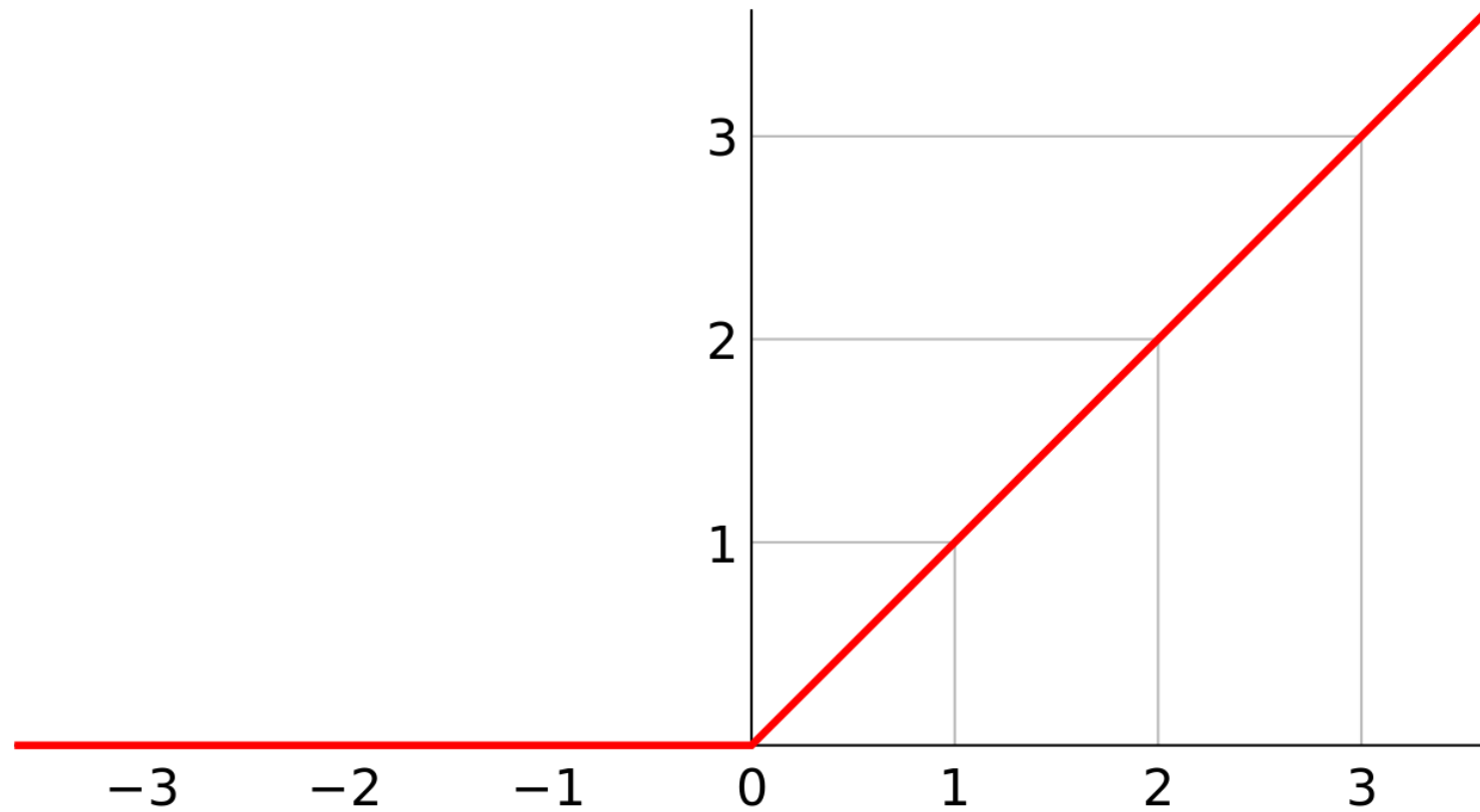
m **message** function



a **aggregation** function: sum

u **update** function

ReLU



Source: https://commons.wikimedia.org/wiki/File:Ramp_function.svg



Example

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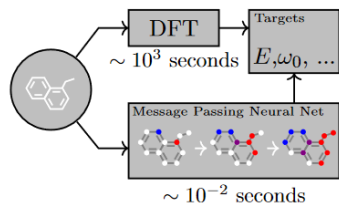


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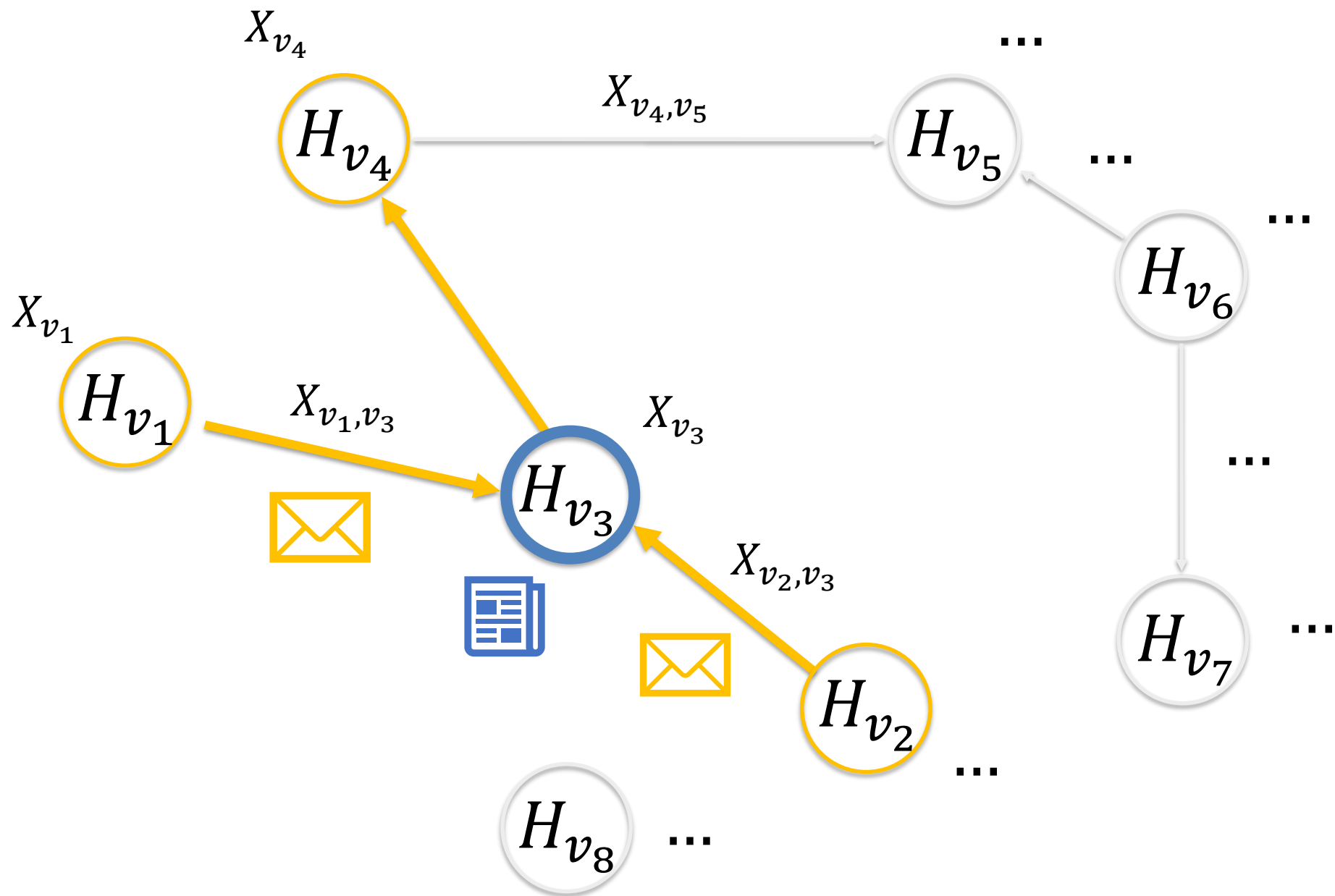
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Graph Neural Networks

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